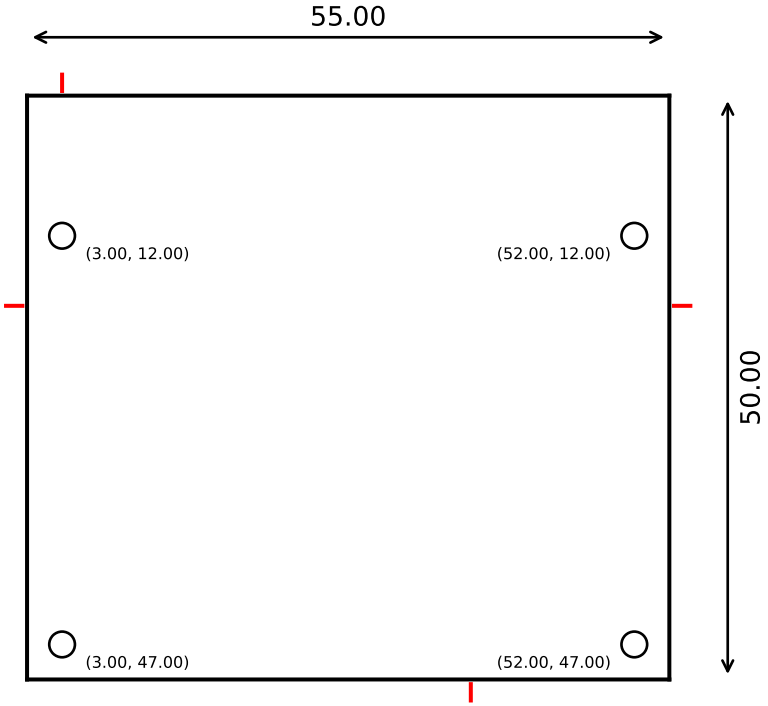
display-adapter — fabrication drawing



Finished thickness: 1.6 mm +/-10%

4 layers, FR-4

35 um copper, all layers

ENIG

Green mask / white silk

IPC Class 2

Electrical test: 100%

Red: bridge suggestions

Layer order: F.Cu / In1.Cu / In2.Cu / B.Cu. Routed Edge.Cuts and drill files are authoritative. Four-layer dielectrics: 0.14 / 1.20 / 0.14 mm, nominal finished 1.6 mm. GIGA concave milling radii: 0.9 mm.

Minimum track 0.15 mm; copper clearance 0.125 mm. Through via drill 0.25 mm, diameter 0.55 mm, annular ring >=0.15 mm. Exposed vias, no via-in-pad. Zero source mask expansion; processed fine-pitch webs >=0.10 mm. Component-hole rings >=0.30 mm.

AISLER bare boards only; ENIG on both boards. Use the ODB++ upload ZIP, including the aisler-bridges document layer. Keep the closed profile unchanged. PTH and NPTH drills are separate. No assembly or stencil is ordered.

Origin in Gerbers, drills and placement: X=0, Y=lower board edge; positive Y upward. Drawing coordinates above use KiCad top-left Y-down for mechanical reference.

Optional paste reference only (no stencil ordered): 0.15 mm at FH28 and 0.10–0.12 mm at 0402/TSSOP parts. Hirose apertures are reduced in the paste layers. No substitutions for connector suffixes or header tail lengths.

display-adapter — assembly instructions

Use assembly-F.pdf and assembly-B.pdf for reference designators, pin numbers and component orientation. Bottom drawing is mirrored to show the actual bottom view. Native KiCad placement rotations are retained in placement.csv.

All components in bom-all.csv are FIT and installed separately by the builder. bom-smd.csv and placement.csv are placement aids only. HAND denotes through-hole installation; no assembly service is included.

Inspect every 0.5 mm connector contact for bridges, insufficient solder and coplanarity. Check soldermask webs and that no solder or exposed via intrudes into the DF23 contact-band keepout.

J1 DF23C-50DP-0.5V(92) is BOTTOM mounted. Viewed from the PCB top: pin 1 is lower left of the contact array, pin 2 above it, pin 49 lower right. Cross-check Smartwin rear-view Detail A before mating.

J2 FH28H is TOP mounted, mouth facing the upper board edge, pin 1 at the right end viewed from the top. Fit the cable with the specified same-side contact orientation.

U1 TRACO TSR 2-2433 is HAND soldered on TOP. Pin 1 = 24V in, pin 2 = GND, pin 3 = 3.3V out. Follow the pin-1 marking; the 12V converter is NOT interchangeable.

After soldering U1 and U2, clip their leads to at most 0.8 mm below the PCB, preserving the solder joints. Untrimmed module leads protrude below the nominal display mating plane. The STEP assembly model shows trimmed leads.

U2 is TSR 1-2450 (5V); U3 is TPS7A91 (3.296V). Solder the WSON exposed pad with paste and controlled hot air/reflow. No via filling is required. Never swap converter types.

Support the PCB on four M2 insulating mounts and clamp the FFC/the ribbon to the enclosure. The display flex and DF23 must carry no cable or PCB weight. Maintain nominal 1.5 mm mating-board spacing at DF23, independent of the 1.6 mm PCB thickness.

First-board power, timing, touch and supply-sequencing tests are listed in README.md. They have not been performed.